Quantitative Finance

ISSN: 1469-7688 (Print) 1469-7696 (Online) Journal homepage: www.tandfonline.com/journals/rquf20

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To cite this article: Oguz Ersan & Montasser Ghachem (2025) A methodological approach to the computational problems in the estimation of adjusted PIN model, Quantitative Finance, 25:7, 1133-1145, DOI: 10.1080/14697688.2025.2515929

To link to this article: https://doi.org/10.1080/14697688.2025.2515929

Published online: 02 Jul 2025.

Quantitative Finance, 2025
Vol. 25, No. 7, 1133–1145, https://doi.org/10.1080/14697688.2025.2515929

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A methodological approach to the computational problems in the estimation of adjusted PIN model

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(*Received 29 December 2024; accepted 22 May 2025*)

It is well documented that computational problems may lead to large biases in the estimation of probability of informed trading (PIN) models. The complexity of the AdjPIN model [Duarte, J. and Young, L., Why is PIN priced? *J. Financ. Econ.*, 2009, **91**, 119–138.], an extension of the conventional PIN model, exacerbates further these computational issues due to its larger parameter set. We introduce a dual approach to improve estimation reliability: a logarithmic factorization of the likelihood function and a strategic algorithm for generating initial parameter sets. The logarithmic factorization addresses floating point exceptions and numerical instability, while the algorithm significantly reduces the likelihood of converging to local maxima. We show that our methodology outperforms existing best practices and it enables accurate estimation of the AdjPIN model. We, therefore, strongly suggest its use in future studies.

Keywords: Adjusted probability of informed trading; AdjPIN; Cluster analysis; Maximum-likelihood estimation; Information asymmetry; Expectation–Maximization algorithm

JEL Classification: C13, C38, G14, G17

1. Introduction

Probability of Informed Trading (PIN) has been one of the most widely used measures of informed trading since its introduction in Easley *et al.* (1996) based on the theoretical background in Easley and O'hara (1987) and Easley and O'hara (1992). The PIN model assumes two different types of trades (informed and uninformed) within three different types of days (days with no-information event, days with good (bad) information event). While uninformed traders in the market are active in all trading days, informed buyers (sellers) are active only on days with good (bad) information event days. The model makes three additional assumptions: (i) trading days are independent from one another, (ii) information events occur outside trading hours, and (iii) at most one information event occurs per day. Duarte and Young (2009) challenge the assumption that trading is only performed by uninformed liquidity traders and informed traders. The authors document the excessively high correlation between the buys and sells in real data, and point out the need to consider for the possibility of liquidity shocks to both the buy and sell side. In light of these observations, their paper introduces an extended model called the Adjusted PIN model (AdjPIN model henceforth) that incorporates symmetric liquidity shocks. According to the AdjPIN model, there can be six types of days. Specifically, each of the three types of days assumed in PIN model (no-event, good event, bad event) has either a symmetric order-flow shock or not. Based on the AdjPIN model estimates, two measures are calculated, i.e. ADJPIN (an adjusted PIN measure) and PSOS (probability of symmetric order-flow shock). Duarte and Young (2009) show that the information asymmetry component, represented by AdjPIN, is not priced, while the liquidity component, represented by PSOS, is priced. This finding challenged earlier findings of Easley *et al.* (2002) and Easley and O'hara (2004). The AdjPIN model has been used extensively in the last decade, making the referred paper the most cited paper in this branch of literature, immediately after the seminal papers introducing the PIN model by Easley, O'Hara and coauthors.

Even though PIN (and AdjPIN) model has been broadly, and extensively implemented in the literature, its estimation remains prone to significant computation biases unless it is accompanied with remedial computational methods—especially with the heavy trading activity in today's financial markets. These computational issues are (i) floating

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point exception problem arising due to large numbers (of trades) in the power term in likelihood function and (ii) frequently reaching local rather than global maxima in the maximum likelihood estimation (MLE). The literature suggests methods to circumvent biases that may result due to these issues in the original PIN model. The floating point exception problem is successfully alleviated by factorizations (logarithmic transformations) of the PIN likelihood function (Lin and Ke 2011, Ersan 2016). The problem of local maxima is resolved by priming the MLE with strategically determined and/or multiple sets of initial parameters (Yan and Zhang 2012, Gan *et al.* 2015, Ersan and Alici 2016). Ersan and Alici (2016) show that running the MLE five times with strategically determined initial sets and via the use of the factorization of Lin and Ke (2011) leads to unbiased estimates of the five intermediate parameters as well as of PIN. Given that the AdjPIN model has a larger set of parameters (10 parameters as opposed to 5 in PIN model), its estimation is more likely to suffer from the aforementioned computational issues. Still, the literature has not examined the magnitude, and the impact of such biases on the model estimates, nor has it provided the necessary computational methods to obtain unbiased estimates. Consequently, it is highly probable that a significant part of the literature implementing the AdjPIN model relied on biased estimates.

In this paper, we attempt to fill this gap in the literature and provide remedial solutions for the estimation of the AdjPIN model. First, we derive a factorization for the likelihood function of the AdjPIN model that is in line with Ersan (2016) factorization for the PIN model. Next, we suggest an algorithm that strategically generates initial parameter sets to be used in priming the maximum-likelihood estimation of the AdjPIN model. Utilizing the recently introduced R package, PINstimation, we compare estimates of the AdjPIN model using the methods suggested in this paper and estimates obtained not using them. We show that our suggested methods provide substantially improved estimates, as the estimates for 10 intermediate parameters, as well as, for AdjPIN, and PSOS suffer from marginal biases for the vast majority of estimations. Maximizations of the original AdjPIN model likelihood function (without the suggested log transformation) do not converge when maximum number of daily buys or sells exceeds around 140. This suggests that the use of log-likelihood function is inevitable in today's financial markets. Next, the use of the new method for the generation of initial parameter sets (GE method) leads to a substantial increase in the accuracy of estimates. In fact, using 20 initial sets generated by the GE method produces better estimates than using 250 randomly generated sets. We also provide extensive supporting evidence for the relevance of our method, by estimating 20 different variants of the (restricted) AdjPIN model alongside the unrestricted model. In light of these results, we suggest the combined use of 20 initial sets generated using the GE method, and the factorization of the likelihood function in the estimation of the AdjPIN model.

To our knowledge, this paper is the first to provide an effective and holistic set of computational methods to mitigate potentially large biases in the estimation of the AdjPIN model.† Future studies relying on the AdjPIN model may significantly benefit from the introduced methods of this paper. Both academicians and practitioners can easily implement the introduced computation methods via the use of specific functions and arguments in PINstimation package. We expect that the suggested methods, as accompanied with the coding utilities, will enable more precise estimation of informed trading and broader use of the AdjPIN model in the future.

2. COMPUTATIONAL CHALLENGES IN THE MODEL ESTIMATION

2.1. *Factorization of the AdjPIN model likelihood function*

Due to heavy trading in financial markets, number of daily buys and sells, that are used in the power terms in the likelihood functions of PIN and AdjPIN (equation 1) models, are frequently too large. In contrast, statistical programs can calculate in limited ranges. For example, the R software computing $\exp(x)$ leads to an overflow problem whenever x exceeds 709. Therefore, the maximum likelihood estimation of the function reaches a maximum point with a narrow set of solutions when the numbers of buys and sells exceed certain limits. Lin and Ke (2011) treat the phenomenon in detail, define the problem rigorously and provide a remedial solution for the estimation of PIN model. In the following steps, we present a logarithmic transformation of the AdjPIN model likelihood function, preventing the overflow problem and the associated biases.

Assuming that trades are distributed according to the assumptions of the adjusted PIN model by Duarte and Young (2009), i.e. according to a mixture model where buys and sells for each day type follow a bivariate Poisson distribution, the likelihood of observing B buyer-initiated trades (or buys) and S seller-initiated trades (or sells) on a trading day is given by equation (1).

$$
\begin{aligned}
L\left((B,S)|\boldsymbol{\theta}\right) &= (1-\alpha)(1-\theta)e^{-\varepsilon_b}\frac{\varepsilon_b^B}{B!}e^{-\varepsilon_s}\frac{\varepsilon_s^S}{S!} \\
&\quad + (1-\alpha)\theta e^{-\varepsilon_b-\Delta_b}\frac{(\varepsilon_b+\Delta_b)^B}{B!}e^{-\varepsilon_s-\Delta_s} \\
&\quad \times \frac{(\varepsilon_s+\Delta_s)^S}{S!} + \alpha(1-\delta)(1-\theta')e^{-\varepsilon_b-\mu_b} \\
&\quad \times \frac{(\varepsilon_b+\mu_b)^B}{B!}e^{-\varepsilon_s}\frac{\varepsilon_s^S}{S!} + \alpha(1-\delta)\theta' e^{-\varepsilon_b-\mu_b-\Delta_b}
\end{aligned}
$$

† One exception is the paper of Cheng and Lai (2021). While the paper discusses the computational issues, it has several limitations in addressing them. First, they examine only a restricted version of the adjPIN model where the probability of liquidity shocks is the same for both no information and information days ($\theta = \theta'$). Second, the number of suggested initial parameter sets—when extended for the unrestricted model—is large, thus time inefficient, without preventing the large estimation biases. Third, the paper relies on normal distribution approximation of the Poisson distribution, which (1) is unjustified in the presence of a factorization of the actual Poisson mixture distribution and (2) can be biased when the level of buys and sells are small, which is the case for infrequently traded stocks.

$$\times \frac{(\varepsilon_b+\mu_b+\Delta_b)^B}{B!}e^{-\varepsilon_s-\Delta_s}\frac{(\varepsilon_s+\Delta_s)^S}{S!}$$
$$+\alpha\delta(1-\theta')e^{-\varepsilon_b}\frac{\varepsilon_b^B}{B!}e^{-\varepsilon_s-\mu_s}\frac{(\varepsilon_s+\mu_s)^S}{S!}$$
$$+\alpha\delta\theta' e^{-\varepsilon_b-\Delta_b}\frac{(\varepsilon_b+\Delta_b)^B}{B!}e^{-\varepsilon_s-\mu_s-\Delta_s}$$
$$\times \frac{(\varepsilon_s+\mu_s+\Delta_s)^S}{S!}, \quad (1)$$

where α is the occurrence probability of an information event; δ is the probability that the information event is a bad event; $\theta(\theta')$ is the probability of a liquidity shock in no-information (information) days; $\mu_b(\mu_s)$ is the informed trading intensity on the buy (sell) side; $\Delta_b(\Delta_s)$ is the intensity of the liquidity shock on the buy (sell) side; and ε_b and ε_s are uninformed trading intensities on the buy and sell sides, respectively. The likelihood function on a day is the summation of six likelihood values on that day, given the daily numbers of buys and sells for a timespan (usually a quarter). The likelihood values on the right-hand side of equation (1) represent the likelihood that the day is (1) a day with no information event and no liquidity shock, (2) a day with no information event and a liquidity shock, (3) a day with a good information event and no liquidity shock, (4) a day with a good information event and a liquidity shock, (5) a day with a bad information event and no liquidity shock, (6) a day with a bad information event and a liquidity shock, respectively.

For a time period of I days, the joint likelihood of observing a set of daily buys and sells, $M=(B_i,S_i)_{i=1}^I$ is presented as in equation (2).

$$L(M|\theta)=\prod_{i=1}^{I}L((B_i,S_i)|\boldsymbol{\theta}) \quad (2)$$

Factoring out the common terms in equation (1), we obtain

$$L((B,S)|\boldsymbol{\theta})=\frac{e^{-\varepsilon_b-\varepsilon_s}\varepsilon_b^B\varepsilon_s^S}{B!S!}$$
$$\times\Big[(1-\alpha)(1-\theta)+(1-\alpha)\theta e^{-\Delta_b}$$
$$\times\left(1+\frac{\Delta_b}{\varepsilon_b}\right)^B e^{-\Delta_s}\left(1+\frac{\Delta_s}{\varepsilon_s}\right)^S$$
$$+\alpha(1-\delta)(1-\theta')e^{-\mu_b}\left(1+\frac{\mu_b}{\varepsilon_b}\right)^B$$
$$+\alpha(1-\delta)\theta' e^{-\mu_b-\Delta_b}$$
$$\times\left(1+\frac{\mu_b+\Delta_b}{\varepsilon_b}\right)^B e^{-\Delta_s}\left(1+\frac{\Delta_s}{\varepsilon_s}\right)^S$$
$$+\alpha\delta(1-\theta')e^{-\mu_s}\left(1+\frac{\mu_s}{\varepsilon_s}\right)^S+\alpha\delta\theta' e^{-\Delta_b}$$
$$\times\left(1+\frac{\Delta_b}{\varepsilon_b}\right)^B e^{-\mu_s-\Delta_s}\left(1+\frac{\mu_s+\Delta_s}{\varepsilon_s}\right)^S\Big] \quad (3)$$

Defining

$$e_0=0 \quad e_1=-\Delta_b-\Delta_s+B\log\left(1+\frac{\Delta_b}{\varepsilon_b}\right)$$
$$+S\log\left(1+\frac{\Delta_s}{\varepsilon_s}\right)$$
$$e_2=-\mu_b+B\log\left(1+\frac{\mu_b}{\varepsilon_b}\right) \quad e_3=-\mu_b-\Delta_b-\Delta_s$$
$$+B\log\left(1+\frac{\mu_b}{\varepsilon_b}+\frac{\Delta_b}{\varepsilon_b}\right)+S\log\left(1+\frac{\Delta_s}{\varepsilon_s}\right)$$
$$e_4=-\mu_s+S\log\left(1+\frac{\mu_s}{\varepsilon_s}\right) \quad e_5=-\mu_s-\Delta_s-\Delta_b$$
$$+B\log\left(1+\frac{\Delta_b}{\varepsilon_b}\right)+S\log\left(1+\frac{\mu_s}{\varepsilon_s}+\frac{\Delta_s}{\varepsilon_s}\right)$$

We can rewrite $L((B,S)|\boldsymbol{\theta})$ as

$$L((B,S)|\boldsymbol{\theta})=\frac{e^{-\varepsilon_b-\varepsilon_s}\varepsilon_b^B\varepsilon_s^S}{B!S!}[(1-\alpha)(1-\theta)\exp e_0$$
$$+(1-\alpha)\theta\exp e_1+\alpha(1-\delta)\left(1-\theta'\right)\exp e_2$$
$$+\alpha(1-\delta)\theta'\exp e_3+\alpha\delta\left(1-\theta'\right)\exp e_4$$
$$+\alpha\delta\theta'\exp e_5]. \quad (4)$$

Defining $e_{\max}=\max(e_0,e_1,e_2,e_3,e_4,e_5)$, and multiplying and dividing $L((B,S)|\boldsymbol{\theta})$ by $e^{+e_{\max}}$:

$$L((B,S)|\boldsymbol{\theta})=\frac{e^{-\varepsilon_b-\varepsilon_s}\varepsilon_b^B\varepsilon_s^S}{B!S!}e^{+e_{\max}}[(1-\alpha)(1-\theta)$$
$$\times\exp(e_0-e_{\max})+(1-\alpha)\theta\exp(e_1-e_{\max})$$
$$+\alpha(1-\delta)(1-\theta')\exp(e_2-e_{\max})$$
$$+\alpha(1-\delta)\theta'\exp(e_3-e_{\max})$$
$$+\alpha\delta(1-\theta')\exp(e_4-e_{\max})$$
$$+\alpha\delta\theta'\exp(e_5-e_{\max})] \quad (5)$$

Finally, taking the log of the likelihood function, we get

$$\log[L((B,S)|\boldsymbol{\theta})]=\log[(1-\alpha)(1-\theta)\exp(e_0-e_{\max})$$
$$+(1-\alpha)\theta\exp(e_1-e_{\max})$$
$$+\alpha(1-\delta)\left(1-\theta'\right)\exp(e_2-e_{\max})$$
$$+\alpha(1-\delta)\theta'\exp(e_3-e_{\max})$$
$$+\alpha\delta\left(1-\theta'\right)\exp(e_4-e_{\max})$$
$$+\alpha\delta\theta'\exp(e_5-e_{\max})]+B\log\varepsilon_b$$
$$+S\log\varepsilon_s-(\varepsilon_b+\varepsilon_s)$$
$$+e_{\max}-\log(B!S!). \quad (6)$$

Similar to equation (2), the joint log-likelihood of observing a set of daily buys and sells for a time period of I days, $M=(B_i,S_i)_{i=1}^I$ is presented as in equation (7).

$$\log[L(M|\boldsymbol{\theta})]=\sum_{i=1}^{I}\log[L((B_i,S_i)|\boldsymbol{\theta})] \quad (7)$$

2.2. Determination of initial parameter sets

The estimation of PIN and AdjPIN models can frequently result in boundary solutions for the probability terms (i.e. $\alpha, \delta, \theta, \theta'$) in the estimated parameter sets. Consequently, the estimation procedure can land on local maxima, failing, thereby, to reach the global maxima of the likelihood function. Yan and Zhang (2012) report that more than one-fourth of the estimations in a 2004 data for 1601 stocks result in boundary solutions in the estimation of PIN model via LK factorization. The excessive number of boundary appearance signals a frequent problem of reaching to a local rather than global maxima.† To address the issue, Yan and Zhang (2012) suggest an algorithm (YZ algorithm) for generating multiple (up to 125) initial parameters sets. For each initial parameter set, the log-likelihood function, as factorized by Lin and Ke (2011), is maximized to obtain a corresponding optimal estimate. The optimal estimate with the highest likelihood value among all is selected. In subsequent works, Gan *et al.* (2015) suggest the use of a clustering algorithm in determining a single and efficient set of parameters; and Ersan and Alici (2016) present another clustering algorithm (EA algorithm) to run the function with small number of sets (use of 5 sets are shown to be bias-free) selecting the run with the highest likelihood value. The problem of boundary solutions—and more generally the failure to reach the global maxima—should be more pronounced in the case of the AdjPIN model due to the larger set of parameters to be estimated.‡

In this section, we present an algorithm for the determination of multiple initial parameter sets to prime the maximization of the log-likelihood function for the AdjPIN model derived in section 2.1. Algorithms for determining initial parameter sets manipulate the data to derive estimates of the model parameters using, for instance, clustering techniques (Ersan and Alici 2016, Gan *et al.* 2015) or sorting, and averaging techniques (Cheng and Lai 2021, Yan and Zhang 2012). We develop here a novel approach to generate such estimates, that is inspired by the EM algorithm.† The spirit of the algorithm can be exposed as follows: assume we are given exogenous (theoretical) distributions for buys and sells for the different day types. Using these distributions, we aim to assign a day type for each daily observation in our data. To do so, we compute the probability that the daily observation (pair of buys and sells) is generated by the distribution relative to each of the 6 day types (defined in table 1) and assign to it the day type associated with the highest probability. Once all observations are assigned day types, we compute the average buys and sells associated with each day type. Then, by leveraging theoretical relationships between the parameters of different distributions, we estimate the model parameters. This overview does not delve into the specifics of obtaining the exogenous distributions, practically distributing daily observations, or the details of computing the model parameters. These details will be elucidated below.

More specifically, steps 1–3 detail the algorithm for obtaining one initial parameter set, step 4 exposes the details of its extension to obtain multiple initial parameter sets, while step 5 describes the process of generating a predefined number of initial parameter sets.

Step 1: *Generate estimates for the distribution means of the AdjPIN model* We aim here to derive estimates for the means of the Poisson distribution for each day type in the AdjPIN model. We denote the estimate of the mean parameter x by $(x)^e$. For instance, we denote the estimate of the mean of buys in no-information days without liquidity shocks by $(\varepsilon_b)^e$.

Our strategy of deriving these estimates rely on the observations about the distribution means of buys and sells as shown in table 1. Note that means of buys take four distinct values $(\varepsilon_b, \varepsilon_b + \mu_b, \varepsilon_b + \Delta_b, \varepsilon_b + \mu_b + \Delta_b)$. Since all means are assumed to be positive, $\varepsilon_b(\varepsilon_b + \mu_b + \Delta_b)$ is evidently the smallest (largest) mean; while it is not clear how to rank $\varepsilon_b + \mu_b$ and $\varepsilon_b + \Delta_b$. Checking the means of sells associated with these means allows us to conclude that the means of sells associated with $\varepsilon_b + \mu_b$ is ε_s, and is therefore smaller than the means of sells associated with $\varepsilon_b + \Delta_b$ that is either $\varepsilon_s + \Delta_s$ or $\varepsilon_s + \mu_s + \Delta_s$. A similar reasoning allows us to draw conclusions on the ranking of the means of sells. Keeping these observations in mind, we proceed with the following steps:

(1) Cluster the data by buys into four clusters. Using the hierarchical agglomerative clustering algorithm, we cluster the dataset by buys into four clusters. For each of these four clusters, we compute the average buys and the average sells. The minimum average buys and the maximum average buys are estimates of $\varepsilon_b : (\varepsilon_b)^e$ and of $\varepsilon_b + \mu_b + \Delta_b : (\varepsilon_b + \mu_b + \Delta_b)^e$, respectively. Among the two remaining clusters, the average buys associated with the largest and smallest average sells are the estimates of $\varepsilon_b + \Delta_b : (\varepsilon_b + \Delta_b)^e$ and of $\varepsilon_b + \mu_b : (\varepsilon_b + \mu_b)^e$, respectively.

(2) Cluster the data by sells into four clusters. Using the hierarchical agglomerative clustering algorithm, we cluster the dataset by sells into four clusters. For each of these four clusters, we compute the average buys and the average sells. The minimum average sells, and the maximum average sells are estimates of $\varepsilon_s : (\varepsilon_s)^e$ and of $\varepsilon_s + \mu_s + \Delta_s : (\varepsilon_s + \mu_s + \Delta_s)^e$, respectively. Among the two remaining clusters, the average sells associated with the largest and smallest average buys are the estimates of $\varepsilon_s + \Delta_s : (\varepsilon_s + \Delta_s)^e$ and of $\varepsilon_s + \mu_s : (\varepsilon_s + \mu_s)^e$, respectively.

Using the estimates obtained above, we can build a bivariate Poisson distribution for each day type T_k for $k \in \{1, \ldots, 6\}$, and whose means are given by the average rates B_k^e, and S_k^e as reported in table 2.

† Boundary solutions are defined as the estimates close to 0 (< 0.02) or 1 (> 0.98) for the probability terms. Especially for the probability of information event occurrence (alpha), frequent appearance of boundary solutions is a signal for biased estimation since we do not expect to see that none (all) of the days in a period has (have) a private information event (Ersan and Alici 2016, Yan and Zhang 2012).

‡ In section 3, we document large biases in the AdJPIN model estimation, using one set of initial parameters.

† The Expectation-Maximization (EM) algorithm is a powerful iterative technique commonly used for estimating parameters in statistical models, particularly in the context of mixture models (Dempster *et al.* 1977). It comprises two key steps: the E-step, where latent variables are estimated based on current parameter values, and the M-step, where model parameters are updated to maximize the likelihood of the observed data. During the assignment step of the EM algorithm, data observations are assigned to different distributions based on the posterior probabilities of belonging to each distribution, computed in the E-step. See Ghachem and Ersan (2025) for the adaptation of the EM algorithm to PIN models.

Table 1. Day-type definitions for the AdjPIN model.

Day type			Event day			
Event type	No-event day		Good event		Bad event	
Liquidity shock	N	Y	N	Y	N	Y
$B_i \sim Po(\lambda_b) : \lambda_b$	ε_b	$\varepsilon_b + \Delta_b$	$\varepsilon_b + \mu_b$	$\varepsilon_b + \mu_b + \Delta_b$	ε_b	$\varepsilon_b + \Delta_b$
$S_i \sim Po(\lambda_s) : \lambda_s$	ε_s	$\varepsilon_s + \Delta_s$	ε_s	$\varepsilon_s + \Delta_s$	$\varepsilon_s + \mu_s$	$\varepsilon_s + \mu_s + \Delta_s$
Day type	T_1	T_2	T_3	T_4	T_5	T_6

Table 2. Estimates of distribution means for buys and sells per day type.

Day type T_k	T_1	T_2	T_3	T_4	T_5	T_6
B_k^e	$(\varepsilon_b)^e$	$(\varepsilon_b + \Delta_b)^e$	$(\varepsilon_b + \mu_b)^e$	$(\varepsilon_b + \mu_b + \Delta_b)^e$	$(\varepsilon_b)^e$	$(\varepsilon_b + \Delta_b)^e$
S_k^e	$(\varepsilon_s)^e$	$(\varepsilon_s + \Delta_s)^e$	$(\varepsilon_s)^e$	$(\varepsilon_s + \Delta_s)^e$	$(\varepsilon_s + \mu_s)^e$	$(\varepsilon_s + \mu_s + \Delta_s)^e$

Step 2: *Assigning day types to observations* In this step, we aim to assign to each daily observation x_t, a pair of buys and sells (B_t, S_t) one of the day types $(T_1, \ldots, T_6)$. To do that, we compute, for each day type, the probability that the pairs of buys and sells are generated by the bivariate Poisson distribution associated with that type. The observation is then attached to the day type associated with the highest probability. Technically, for each daily observation $x_t : (B_t, S_t)$ and a day type T_k, we calculate the aforementioned probability $\mathbb{P}[x_t \sim T_k]$ as follows:

$$\mathbb{P}[x_t \sim T_k] = P[X_k^b = B_t] \times P\left[X_k^s = S_t\right], \tag{8}$$

where X_k^b and X_k^s are two Poisson-distributed random variables associated with day type T_k, with means B_k^e, and S_k^e respectively. A problem, however, occurs when for some observation x_t, all probabilities $\mathbb{P}[x_t \sim T_k], \forall k$ are equal to 0, so it is not possible to select the day type with the highest probability.† Since $P[X_b = B_t]$ is decreasing in $\left|B_k^e - B_t\right|$, and that $P[X_s = S_t]$ is decreasing in $\left|S_k^e - S_t\right|$; we have, therefore, opted to approximate the likelihood that observation x_t is generated by the bivariate Poisson distribution associated with day type T_k by a likelihood score $\mathcal{L}[x_t \sim T_k]$, which is the negative of Euclidean distance between the rates (B_k^e, S_k^e), and (B_t, S_t), which is immune to aforementioned problem.

$$\mathcal{L}[x_t \sim T_k] = -\sqrt{\left(B_k^e - B_t\right)^2 + \left(S_k^e - S_t\right)^2} \tag{9}$$

This multiplication by -1 is motivated by the requirement that the likelihood score ought to increase when the distance between rates becomes smaller. We apply this procedure for all daily observations x_t with $t = 1 \ldots n$, at the end of which, each of them is attached to a day type T_k, for some k. Now, for each day type T_k, we associate the following statistics: n_k the number of observations assigned to the day type T_k, $\bar{B}_k$ their average buys, and $\bar{S}_k$ their average sells. If a day type T_j has no observation attached to it, then $n_j = \bar{B}_j = \bar{S}_j = 0$.

Step 3—*Computing initial parameter sets* Let $n_k, \bar{B}_k$, and $\bar{S}_k$, the number of days, the average of buyer-initiated trades and seller-initiated trades for day type T_k as defined in table 3. Following the model assumptions, we calculate the model parameters as follows:

Table 3. Empirical values of days, buys, and sells per cluster.

Cluster	C_1	C_2	C_3	C_4	C_5	C_6
Days	n_1	n_2	n_3	n_4	n_5	n_6
Average Buys	$\bar{B}_1$	$\bar{B}_2$	$\bar{B}_3$	$\bar{B}_4$	$\bar{B}_5$	$\bar{B}_6$
Average Sells	$\bar{S}_1$	$\bar{S}_2$	$\bar{S}_3$	$\bar{S}_4$	$\bar{S}_5$	$\bar{S}_6$

$$\hat{\alpha} = \frac{n_3 + n_4 + n_5 + n_6}{n} \qquad \hat{\delta} = \frac{n_5 + n_6}{(n_3 + n_4 + n_5 + n_6)}$$

$$\hat{\theta} = \frac{n_2}{n_1 + n_2} \qquad \hat{\theta}' = \frac{n_4 + n_6}{(n_3 + n_4 + n_5 + n_6)}$$

$$\hat{\varepsilon}_b = \max\left(\bar{B}_1, \bar{B}_5\right) \qquad \hat{\varepsilon}_s = \max\left(\bar{S}_1, \bar{S}_3\right)$$

$$\hat{\Delta}_b = \max\left(\bar{B}_2 - \hat{\varepsilon}_b, \bar{B}_4 - \hat{\varepsilon}_b, 0\right)$$

$$\hat{\Delta}_s = \max\left(\bar{S}_2 - \hat{\varepsilon}_s, \bar{S}_4 - \hat{\varepsilon}_s, 0\right)$$

$$\hat{\mu}_b = \max\left(\bar{B}_3 - \hat{\varepsilon}_b, \bar{B}_4 - \hat{\varepsilon}_b - \hat{\Delta}_b, 0\right)$$

$$\hat{\mu}_s = \max\left(\bar{S}_5 - \hat{\varepsilon}_s, \bar{S}_6 - \hat{\varepsilon}_s - \hat{\Delta}_s, 0\right).$$

If either of $\hat{\varepsilon}_b$ and $\hat{\varepsilon}_s$ is equal to zero, then the estimates ε_b^e and ε_s^e are used. The formulas are based on the theoretical relationships between the distribution means of buys and sells as reported in table 1.‡

Step 4—*Generating multiple initial parameter sets* Applying the steps from 1 to 3 gives us a single initial parameter set. To produce multiple initial parameter sets, we perform step 1

† This occurs when, for a given observation x_t, and for all day types T_k, the computer assigns 0 to either $P[X_k^b = B_t]$ or $P[X_k^b = S_t]$. Such probabilities are theoretically never zero, but are sufficiently low, that the computer gives them this value.

‡ The use of the maximum operator, rather than the average, is empirically motivated. In rare cases, the presence of highly uninformative or anomalous values can introduce downward bias when using the average. By contrast, the maximum operator mitigates this issue, ensuring that such irregularities do not violate the model's non-negativity constraints and do not lead to underestimation of the parameters.

in an extended way. Specifically, we generate multiple distributions using the clustering procedure of the EA algorithm§ (Ersan and Alici 2016).

Our procedure generalizes this clustering approach by partitioning buy (and sell) observations into $4 + x$ clusters, where x is the number of extra clusters—a user-specified parameter of the EA algorithm. Unlike the original method (Step 1), which uses exactly four clusters, this extension allows for finer granularity in the initial clustering.

(a). **Clustering and Sorting**: We first apply hierarchical agglomerative clustering to the buy (sell) observations, generating $4 + x$ clusters ranked by their average buy-side (sell-side) trading volume.

(b). **Recombination into Four Groups**: These clusters are then recombined into **four contiguous groups**, ensuring no non-adjacent clusters are merged. This recombination is mathematically equivalent to selecting three cut-off points between the ordered clusters, where all clusters between two consecutive cut-offs form a single group. The number of valid ways to partition $4 + x$ clusters into four groups is given by the binomial coefficient $\binom{4+x-1}{4-1} = \binom{3+x}{3}$ as derived in Ersan and Alici (2016). Each distinct grouping corresponds to a unique configuration of preliminary parameter estimates.

(c). **Deriving Initial Parameter Sets**: For each grouping, we

- Compute the average buy (or sell) volume per group
- Map these averages to the AdjPIN model's theoretical buy (or sell) parameters (e.g. $B_k^e \in \{\varepsilon_b, \varepsilon_b + \mu_b, \varepsilon_b + \Delta_b, \varepsilon_b + \mu_b + \Delta_b\}$)
- Generate a new initial parameter set by applying steps 2–3.

However, we exclude all duplicated parameter sets, as well as sets where any trade rate estimate for a given day type is zero while the occurrence probability of that day type is positive.† Furthermore, we address edge cases where probability parameters become undefined. For instance, when $\alpha = 1$ (all days are information days) and at least three day types from $\{T_3, T_4, T_5, T_6\}$ are populated, the liquidity shock probability on no-information days (θ) is mathematically indeterminate. In such scenarios, we impose the regularization $\theta = 0.5$ to ensure numerical stability while maintaining theoretical consistency.‡ In section 3.2, we present results for the estimation accuracy of various number of initial sets obtained through aforementioned steps.

Step 5—Preparing a fixed number of initial parameter sets for estimation As described above, for a given number of extra clusters x, steps 1–4 lead to the derivation of up to $\binom{4+x-1}{4-1}$ initial parameter sets. The exact number of initial sets depends on the frequency of non-valid and duplicated initial sets. To evaluate and contrast the accuracy of our method against any benchmark, we set to derive a predetermined number of initial sets. For this purpose, we devise a method to obtain a desired fixed number of initial parameter sets. Assume that we wish to use a number m of initial parameter sets for the estimation of the AdjPIN model. Our method obtains these m sets in the following incremental manner: initially, the number of extra clusters is set to zero ($x = 0$). The corresponding initial parameter set is, then, generated, and stored as *estimation sets*. If the number of these sets reaches or exceeds m, then the process stops. Otherwise, the number of extra clusters is increased by one, the corresponding initial sets are generated and combined with the pre-existing estimation sets. If the new *estimation sets* contain duplicates, they are excluded. Increasing the number of extra clusters continues until the number of sets in *estimation sets*, net of duplicates, reaches or exceeds m. In this case, excessive initial parameter sets (if any) are excluded,§ and the estimation of the AdjPIN model is performed using m initial parameter sets in *estimation sets*. We choose the incremental procedure here because initial sets for low levels of x are typically stronger, while those for higher levels cover more of the parameter space.

In a recent paper, Cheng and Lai (2021) study the initial parameter set determination issue for a simpler version of the AdjPIN model where the probability of a liquidity shock is assumed to be equal in information days and no-information days. The paper suggests and compares three different methods of obtaining initial sets for this restricted model. The first method applies the Yan and Zhang (2012) grid search algorithm to initially obtain $4^7 = 16{,}384$ and select 64 sets randomly among these. Picking 64 sets randomly over 16,384 sets is not representative of the sample space for sure. Thus it is more like a random initial parameter determination. The second method suggested in the paper is an adaptation of Ersan and Alici (2016) initial sets developed for the original

§ Our approach builds on Ersan and Alici (2016), who demonstrate that clustering observations by Absolute Order Imbalance (AOI) yields unbiased PIN estimates, unlike Gan et al. (2015)'s reliance on raw order imbalance. However, we depart from both methods by (i) clustering buys and sells separately into 4 clusters to exploit inherent model parameter rankings (e.g. the lowest-volume buy(sell) cluster estimates correspond to ε_b (ε_s), while the highest estimates correspond to $\varepsilon_b + \mu_b + \Delta_b$ ($\varepsilon_s + \mu_s + \Delta_s$)), avoiding the aggregation pitfalls noted by Ersan and Alici (2016), and (ii) deriving eight preliminary buy/sell parameter estimates using two clustering operations (rather than directly partitioning days into no-/good-/bad-information types in a single clustering operation using AOI). These estimates serve the construction of day-type distribution means (table 2) but are not used directly for final parameter estimation; instead, they enable the day-type assignments in step 2.

† Specifically, if positive (negative) information days exist ($\alpha\delta > 0$ or $\alpha(1-\delta) > 0$), we require $\hat{\mu}_b > 0$ ($\hat{\mu}_s > 0$). Similarly, if liquidity shock days are present ($\theta > 0$ or $\theta' > 0$), we enforce $\hat{\Delta}_b > 0$ and $\hat{\Delta}_s > 0$. This restriction reflects the economic reality that informed trading and liquidity shocks must exhibit non-zero rates when their associated day types have non-zero probabilities.

‡ This regularization is conservative, as it assumes equal likelihood of liquidity shocks in the absence of no-information days, while preserving the model's structural interpretation.

§ We remove excessive sets with random selection. For instance, assume that we start with $x = 0$ and obtain one set. We continue with $x = 1$ and obtain four additional sets. Let us further assume that there are no duplicates and no sets with non-positive parameters. In case we are aiming to reach 10 sets ($m = 10$), we need 5 more sets. We continue with $x = 2$, which provides 10 sets by itself. If there are still no duplicates or non-positive values, 5 out of 10 sets from $x = 2$ are excessive. We randomly select 5 out of 10 sets and combine with the 5 sets in hand (from $x = 0$ or 1). Random selection of excessive sets is reasonable because all of the additional sets are from the same strategy explained in section 2.2, which are supposedly equally likely to contribute.

PIN model. The additional dimension of liquidity shocks in the AdjPIN model is incorporated through a separate clustering step that follows the first clustering step where the no information, good information and bad-information days are grouped. The two-step sequential clustering approach, assuming that the liquidity shocks are conditional on the information events contradicts the model structure leading to inefficient initial parameters. Finally, the third method of initial parameter set determination suggested in the paper significantly relies on grid-search algorithm since all of the included probability terms—as in Yan and Zhang (2012) —are chosen uniformly from the set of $(0.2, 0.4, 0.6, 0.8)$, resulting in 4^3 sets. Initiating with parameters which are exogenous to the data leads to large biases. The paper suggests better comparative performance for the third method. We further extend these sets to $4^4 = 256$ sets since we work with the unrestricted AdjPIN model (four probability parameters instead of three) and report results in the next section.

3. EMPIRICAL TESTS

3.1. *Description of the estimation functions*

The computational investigation of the factorization and initial parameter sets is performed using the R package PINstimation, more specifically with two of its functions: `adjpin()` and `generatedata_adjpin()`. The function `adjpin()` estimates the AdjPIN model of Duarte and Young (2009) and returns the different parameter estimates, the adjusted probability of informed trading (ADJPIN) as well as the probability of symmetric order-flow shock (PSOS). The function has five main arguments:

- `method`: It refers the maximum likelihood estimation method used. It takes two values: `"ML"` referring to the standard MLE and `"ECM"` referring to the expectation-conditional-maximization algorithm. The default method is `"ECM"`.
- `fact`: It indicates whether the factorization of the likelihood function presented in section 2.1 of this paper is applied or not. If `fact=FALSE`, then the original likelihood function (without the log-transformation) is maximized. The default is `fact=TRUE`.
- `initialsets`: It refers to the generation method of initial parameter sets. It takes three values: `"GE"` referring to the method detailed in section 2.2 of this paper, `"CL"` referring to the algorithm in Cheng and Lai (2021), and `"RANDOM"` referring to random generation of initial parameter sets. The default value is `"GE"`.
- `num_init`: It specifies the number of initial sets used in the estimation. The default value is 20.
- `restricted`: A list that allows estimating restricted AdjPIN models by specifying which model parameters are assumed to be equal. For instance, if `restricted = list(theta= TRUE)`, then the AdjPIN model is estimated under the assumption that $\theta = \theta'$.

The function `generatedata_adjpin()` generates a dataset object or a list of dataset objects storing daily aggregate numbers of buys and sells and simulation parameters for the AdjPIN model. The argument `series` specifies the number of datasets generated, the argument `days` specifies the number of days in each generated dataset, the argument `restricted` allows to generate data following restricted AdjPIN models in the same manner described above. To specify the parameters used to generate the datasets, the user can feed them directly through the argument `parameters` or specify the ranges of the different model parameters using the argument `ranges`. If both of these arguments are missing, the model parameters used for data generation are randomly selected from the default ranges specified in the package documentation. For more information about the data simulation process, please refer to Appendix 1.

The steps 1–4 of the algorithm generating the initial parameter sets as detailed in section 2.2 are implemented in the function `initials_adjpin()`. The function `adjpin()` implements step 5, where the argument `num_init` corresponds to the number of initial sets used for the AdjPIN model estimation.

3.2. *Comparative results from estimations*

In this section, we simulate quarterly data series of daily numbers of buys (buyer-initiated trades) and sells (seller-initiated trades). We simulate large number of datasets with various different characteristics. Data simulation is done with the use of the function `generatedata_adjpin()` in PINstimation package. A brief and relevant introduction to the package functions used here is presented in section 3.1 of this paper, while more detailed description can be found in PINstimation package documentation and in Ghachem and Ersan (2023). Using the simulated datasets, we demonstrate that the suggested methods of this paper substantially reduces the biases, otherwise significant, in the absence of the methods.

Table 4 presents the estimation results (mean absolute errors, MAEs) when we do not use the factorization (`fact=FALSE`). We simulate 2500 datasets of 10 types based on total trade intensity. More specifically, for each i in the set $(110, 115, \ldots, 150, 155)$, we simulate 250 sets with maximum daily buys or sells to be between $i + 1$ and $i + 5$. This enables us to see the effect of large numbers' inclusion in the likelihood function. First two columns (columns 3–12) in table 5 present the MAEs for ADJPIN and PSOS (ten intermediate parameters). While absolute errors for probability parameters (columns 1–6) are in level difference (estimate – actual), the ones for rate parameters are in percentage difference (estimate/ actual–1). We use 20 initial sets derived through the method suggested in section 2.2 (GE method). The average number of convergent maximizations over 20 initial sets is 19.9 when the maximum number of buys or sells in the data is in the range of 110–115 and minimally decreases to 19.4 when the range is 130–135. Beyond this range, convergence rates sharply decrease. Specifically, the rate is 9.19/20 for the range 135–140 while it is as low as 0.34/20 for the 140–145 range and 0.02/20 for the 145–150 range. Finally, for all datasets with maximum trading intensity exceeding 150, there is no convergence in any maximization. Convergence rates are supplemented by the variable 'success' provided in the last column. In the estimation of any single dataset, success takes the value 1 if, at least one out of 20 likelihood maximizations converges, and 0 otherwise. The success rate is 100% when the maximum number in the data is below 135, while it is only 1.2% for the range 145–150. An essential result reported in the table regards accuracy: decreasing convergence rates lead to worsening estimation accuracy. For the ranges with low non-zero mean convergence rates (the range between 135 and 150) estimation accuracy decreases substantially. For example, in the range 140–145, PIN and PSOS estimates involve mean errors of 4.08% and 4.52%, much higher than the ones in lower ranges. Moreover, most of the intermediate parameters are estimated with mean errors of more than 20%. The errors are even more exaggerated for the range 145–150: 15.1% and 13.97% for PIN and PSOS and up to 76.67% for model parameters. This is due to the fact that only a few likelihood maximizations converge leading to selection among limited number of estimates. Additionally, as the runs initiated from relatively large values do not converge, the selection is among the ones initiated from smaller values. Overall, the use of the original likelihood function (without log-transformation) is not recommended when the number of buys or sells exceeds 135. Besides, it may have lower estimation accuracy, even in the case of infrequent trading, which depends on the convergence rates.

As a second task, we focus on the accuracy of the estimation of the AdjPIN model using the initial parameter sets generated with the method introduced in section 2.2. In all remaining estimations, as default, we use the log-transformation of the likelihood function. Such use is justified by the results in table 5. In what follows, we conduct two main analyses. The first analysis compares our suggested method's (GE method's) estimation accuracy with three benchmarks. Two of these are preexisting methods in the literature: (i) random initial sets (e.g. Duarte and Young 2009) and (ii) Cheng and Lai (2021) initial sets. The third one comes from a more sophisticated approach that we suggest in generating random sets. In the first analysis, we work with the unrestricted AdjPIN model as suggested in Duarte and Young (2009), with no further restrictions or assumptions. The second analysis focuses on the consistency (robustness) of our initial parameter sets in 20 different versions of the AdjPIN model (restricted models).

Previous papers implementing the AdjPIN model usually disregard the computational challenges inherent to such implementation. In many studies, the estimation method either is not clearly stated or relies on a single set of random parameters. Better implementations use multiple sets of parameters determined randomly. For instance, Duarte and Young (2009), in their paper introducing the model, use 10 random sets in their estimations. Thus, as the first benchmark, we use randomly generated initial sets (*random initial sets*). The generation procedure is similar to earlier works on the original PIN model (e.g. Ersan and Alici 2016). Specifically, the probability variables $(\alpha, \delta, \theta, \theta')$ are independently and uniformly randomly drawn from$(0, 1)$. For the intensity parameters, we apply two restrictions to keep the initial values relevant to the trading intensity in the dataset. Let minB and maxB (minS and maxS) be the smallest and the largest buys (sells) in the data respectively. We generate uniformly randomly ε_b, μ_b, and Δ_b such that $\varepsilon_b \geq$ minB, and $\varepsilon_b + \mu_b + \Delta_b \leq$ maxB. Similarly, we generate uniformly randomly ε_s, μ_s, and Δ_s such that $\varepsilon_s \geq$ minS, and $\varepsilon_s + \mu_s + \Delta_s \leq$ maxS. The motivation of the first restriction is that the day with the lowest buying (selling) intensity would be a day with no good (bad) event and with no liquidity shock, therefore represented by ε_b (ε_s). Similarly, given the model assumptions, the day with the highest buying (selling) activity will likely be a day where uninformed, informed and liquidity buyers (sellers) are present together and represented by $\varepsilon_b + \mu_b + \Delta_b$ $(\varepsilon_s + \mu_s + \Delta_s)$.

Given the fact that the numerical space to be captured with the large parameter set is broad, the performance of the simple random generation procedure might be weak and unstable. Thus we resort to a more sophisticated random algorithm to form a second benchmark (*pseudo-random initial sets*). The value minB (maxB), being the smallest (largest) level of buys, is assumed to be generated by a Poisson distribution with mean $\varepsilon_b(\varepsilon_b + \mu_b + \Delta_b)$, which, therefore, lies in the neighborhood of minB (maxB). Similarly, the value minS (maxS), being the smallest (largest) level of sells, is assumed to be generated by a Poisson distribution with mean $\varepsilon_s(\varepsilon_s + \mu_s + \Delta_s)$, which, therefore, lies in the neighborhood of minS (maxS). We define this neighborhood around a value x as the set of all means m such that the probability that x is generated by a Poisson distribution with mean m is at least 5%—where $x \in \{\text{minB}, \text{maxB}, \text{minS}, \text{maxS}\}$. To determine this neighborhood, we approximate the Poisson distribution with mean m by a normal distribution of mean m, and standard deviation $\sqrt{m}$, and find the lower bound L_x as the solution for $\Phi(\frac{x-m}{\sqrt{m}}) = 5\%$; and the upper bound U_x as the solution for $\Phi(\frac{x-m}{\sqrt{m}}) = 95\%$, where Φ is the distribution function for the standard normal distribution. Once we have obtained intervals, we draw a value b_1 for ε_b uniformly at random from $(\mathrm{L}_{\varepsilon_b}, \mathrm{U}_{\varepsilon_b})$, and a value b_4 from $\varepsilon_b + \mu_b + \Delta_b$ from $(\mathrm{L}_{\varepsilon_b+\mu_b+\Delta_b}, \mathrm{U}_{\varepsilon_b+\mu_b+\Delta_b})$. The difference between b_4 and b_1 is assumed to be an estimate for $\mu_b + \Delta_b$. To generate estimates for μ_b and Δ_b, we draw a value λ uniformly at random from $(0, 1)$, and set $\mu_b = \lambda(b_4 - b1)$ and $\Delta_b = (1 - \lambda)(b_4 - b1)$. Similarly, we draw a value s_1 for ε_s uniformly at random from $(\mathrm{L}_{\varepsilon_s}, \mathrm{U}_{\varepsilon_s})$, and a value s_4 from $\varepsilon_s + \mu_s + \Delta_s$ from $(\mathrm{L}_{\varepsilon_s+\mu_s+\Delta_s}, \mathrm{U}_{\varepsilon_s+\mu_s+\Delta_s})$. The difference between s_4 and s_1 is assumed to be an estimate for $\mu_s + \Delta_s$. To generate estimates for μ_s and Δ_s, we draw a value θ uniformly at random from $(0, 1)$, and set $\mu_s = \theta(s_4 - s1)$ and $\Delta_s = (1 - \theta)(s_4 - s1)$. The probability variables $(\alpha, \delta, \theta, \theta')$ are independently and uniformly randomly drawn from $(0, 1)$.

An alternative method of generating initial parameter sets that relies on a grid-search algorithm similar to Yan and Zhang (2012) is suggested in a recent paper by Cheng and Lai (2021). The paper considers only a restricted AdjPIN model with $\theta = \theta'$, to derive 64 $\left(4^3\right)$ initial parameter sets. To study the accuracy of the method applied to the Duarte and Young (2009) original model (unrestricted model), we extend the method of Cheng and Lai (2021), to derive 256 $\left(4^4\right)$ initial parameter sets, to allow for comparison with our initial parameter sets. Thus we use an extended version of the Cheng

Table 4. Mean absolute errors using the likelihood function without the logarithmic transformation.

	PIN	**PSOS**	α	δ	θ	θ'	ε_b	ε_s	μ_b	μ_s	Δ_b	Δ_s	**Time**	**Conv.**	**Suc.**
111–115	1.05	1.55	5.76	7.02	7.33	5.42	3.33	3.46	12.44	11.59	9.05	9.33	53.15	19.90	1
116–120	0.97	1.37	6.05	6.62	6.38	5.86	3.12	4.62	11.3	11.06	7.91	8.89	52.97	19.87	1
121–125	0.74	1.3	3.48	4.53	4.25	4.12	3.13	3.13	7.12	9.72	7.03	8.25	52.80	19.76	1
126–130	1.04	1.53	4.76	7.16	6.88	5.4	3.07	3.32	11.33	11.38	8.36	9.18	52.96	19.74	1
131–135	0.67	1.1	2.58	4	3.12	2.77	3	2.91	8.78	9.06	6.43	6.38	52.28	19.41	1
136–140	1.34	1.53	8.43	8.29	8.19	7.19	2.97	3.12	12.25	11.02	8.93	10.37	33.45	9.19	0.94
141–145	4.08	4.52	21.11	22.87	22.29	21.82	5.1	7.66	33.32	29.05	17.57	26.88	18.79	0.34	0.15
146–150	15.1	13.97	63.86	17.82	76.67	33.3	13.14	13.65	13.05	46.7	21.68	34.63	17.90	0.02	0.01
151–155	–	–	–	–	–	–	–	–	–	–	–	–	17.76	0	0
156–160	–	–	–	–	–	–	–	–	–	–	–	–	17.79	0	0

Notes: The table presents the mean absolute errors (MAEs) for adjusted probability of informed trading (PIN), probability of symmetric order flow shock (PSOS), and 10 intermediate parameters in the AdjPIN model of Duarte and Young (2009). Original likelihood function (with no factorization) is maximized 20 times with the initial parameters obtained by the GE method in section 2.2. The maximization with the highest likelihood value among all is chosen as optimal. The numbers in the first column refer to the alternative ranges for maximum buys and sells in the data. We simulate 2500 datasets (250 for each of the alternative ranges) based on the assumptions of the AdjPIN model. MAEs for PIN, PSOS as well as the probability variables ($\alpha, \delta, \theta, \theta'$) are in level differences between the estimated and actual values; while MAEs for the rate parameters ($\varepsilon_b, \varepsilon_s, \mu_b, \mu_s, \Delta_b, \Delta_s$) are calculated in percentage differences between the estimated and actual values. All MAE numbers are reported in percentages for readability purpose. For example, in the first row, mean estimation error for PIN is 1.05% (in percentage units), and mean error for ε_b is 3.33% of actual ε_b (in percentage). Runtime is the mean running time in seconds. Last two columns report the mean number of converging maximizations and the mean number of successes. In each estimation, success is one if at least one of the 20 maximizations converged, and 0 otherwise. Data simulation as well as the estimations are conducted through the PINstimation package in R software. Total running time with an Í9 10990K processor is approximately 26 h.

and Lai (2021) method as our last benchmark (*CL initial sets*).

Table 5 reports the MAEs in the same way as in table 4, i.e. PIN, PSOS, and 10 parameters. We simulate 1000 datasets, each representative of quarterly data of daily numbers of buys and sells; generated in accordance to the AdjPIN model assumptions. Table A1 provides descriptive statistics for simulated datasets used in this analysis. The variation in terms of trade intensity, number of buys, and sells, as well as simulation parameters determined in generating the data is substantially high. For instance, minimum (maximum) of the mean number of sells (buys) in datasets is 173.8 (75,618.6). The data generating probability terms α, δ, θ, and θ' all have mean values close to 0.5, while the mean ADJPIN and PSOS are 25% and 34%, respectively.

Panel A of table 5 reports biases for estimations using initial parameter sets derived by our method (GE initial sets) for different numbers of initial parameter sets x where x is a number from the list (1, 5, 10, 20, 50, 100). Panel B reports biases from estimations with random initial sets. Panel C reports biases for the same number of initial parameter sets derived from the more sophisticated algorithm (pseudo-random initial sets). In panels B and C, we repeat the estimations also with 250 sets due to relatively large biases in smaller number of sets. Finally, Panel D reports estimation biases for the 256 sets derived using an extended version of Cheng and Lai (2021) method. We use table 5 to evaluate the comparative performances with respect to (i) number of initial sets and (ii) determination method of initial sets. All panels reflect that the estimation accuracy consistently increases with the number of sets used, that is in line with our expectations and previous studies of the original PIN model (e.g. Ersan and Alici 2016). For example, our method (GE) generates mean absolute errors of 0.2% and 0.46% for PIN and PSOS when we rely on a single set of parameters, strategically determined. These rates are as low as 0.05% and 0.08% when the number of initial sets exceeds 20 sets. A single set of random initial sets (panel B) leads to MAEs in PIN and PSOS as high as 12.60% and 14.91%, while these rates decrease to 3.24% (0.90%) and 3.92% (1.07%) in use of 20 (250) random sets. As Panel C shows, pseudo-random sets that we generate in a more sophisticated way also perform better with the increased number of estimations. MAEs in panel C are substantially lower than the ones in Panel B especially for larger number of sets, implying that an advanced way of generating random sets adds value and stands as a reasonable benchmark for our suggested method. Random sets tend to generate large biases for all intermediate parameters, as well as for PIN and PSOS especially in case of limited number of initial sets. Focusing on panel C, utilizing 20 different sets, MAEs of probability parameters are between 2.98% and 6.26% and MAEs of intensity parameters are between 0.39% and 1.68%. Consequently, MAEs of PIN and PSOS for the same setting (pseudo-random 20) are also high at 1.35% and 1.77% when we consider that the median PIN and PSOS values used in data generation are 25.2% and 34.2%, respectively (table A1).

Increasing the number of random (or pseudo-random) initial sets, even though it consistently reduces the estimation biases, does not constitute a perfect remedy to the biased estimations, despite their long running times.† In contrast, the estimation accuracy of the AdjPIN model is significantly higher using GE sets. The estimation accuracy of GE method improves with the number of sets and this improvement stops at 20 sets. Thus we suggest 20 GE sets as a reasonable default setting. MAE is less than 0.57% for each of the 10 parameters leading to PIN and PSOS with MAEs of only 0.055% and 0.082%. Estimations using 20 sets strategically determined

† The estimation performance of random initial parameter sets remains poor despite the fact that we generate these sets with restrictions. The estimation biases in case of milder restrictions (e.g. replacing the restriction $\varepsilon_b + \mu_b + \Delta_b < maxB$, with $\max(\varepsilon_b, \mu_b, \Delta_b) < maxB$) are even larger.

Table 5. Mean absolute errors with the use of initial parameter sets of different methods.

	PIN	PSOS	α	δ	θ	θ'	ε_b	ε_s	μ_b	μ_s	Δ_b	Δ_s	Time
PANEL A: GE INITIAL SETS													
1	0.200	0.463	0.334	0.647	1.057	0.792	0.31	0.33	0.84	0.74	1.98	2.88	0.3
5	0.075	0.129	0.047	0.108	0.246	0.118	0.28	0.29	0.30	0.30	0.63	0.64	2.3
10	0.064	0.089	0.031	0.088	0.108	0.063	0.28	0.29	0.29	0.29	0.53	0.59	5.1
20	0.055	0.082	0.008	0.032	0.082	0.036	0.28	0.29	0.29	0.29	0.50	0.57	11.4
50	0.055	0.080	0.006	0.017	0.076	0.020	0.28	0.29	0.29	0.29	0.45	0.49	35.5
100	0.054	0.079	0.005	0.015	0.071	0.012	0.28	0.29	0.29	0.29	0.45	0.49	92.4
PANEL B: RANDOM INITIAL SETS													
1	12.60	14.91	27.66	29.09	32.99	31.57	14.38	13.61	29.16	24.56	78.96	78.14	0.4
5	6.84	8.36	15.69	21.66	22.35	22.19	3.27	2.76	4.99	6.31	12.05	23.01	1.9
10	4.99	5.86	11.78	16.37	16.97	16.33	1.20	1.20	1.42	2.16	6.10	9.43	3.8
20	3.24	3.92	7.17	11.63	11.40	12.40	0.83	0.59	0.89	1.84	5.57	4.26	7.6
50	2.02	2.44	4.52	8.18	7.64	7.84	0.36	0.40	0.48	0.48	1.67	0.74	18.8
100	1.24	1.75	2.60	5.57	5.62	5.86	0.31	0.37	0.33	0.34	0.52	0.67	37.7
250	0.90	1.07	1.82	3.58	3.65	3.75	0.30	0.30	0.32	0.31	0.48	0.52	94.3
PANEL C: PSEUDO-RANDOM INITIAL SETS													
1	10.74	14.09	21.05	26.89	29.36	28.08	12.99	12.29	20.73	17.95	238.75	199.93	0.4
5	4.86	5.93	10.51	14.11	15.79	16.20	1.57	1.92	5.04	3.77	18.99	19.44	1.9
10	2.58	3.19	5.59	10.17	10.44	10.87	0.64	0.81	1.24	0.90	4.66	5.99	3.8
20	1.35	1.77	2.98	5.52	6.26	6.06	0.39	0.48	0.45	0.35	1.68	0.81	7.5
50	0.66	0.82	1.33	3.30	3.04	3.43	0.30	0.34	0.32	0.31	0.46	0.60	18.8
100	0.36	0.49	0.70	1.96	1.99	1.87	0.29	0.33	0.30	0.30	0.46	0.57	37.3
250	0.20	0.28	0.34	0.97	0.92	0.99	0.29	0.30	0.29	0.29	0.45	0.49	93.7
PANEL D: CL INITIAL SETS													
256	0.93	1.03	1.93	3.12	3.30	3.27	0.42	0.37	0.72	0.68	1.80	1.01	96.6

Notes: The table presents the mean absolute errors (MAEs) for adjusted probability of informed trading (PIN), probability of symmetric order flow shock (PSOS), and 10 intermediate parameters in the AdjPIN model. GE initial sets refer to the initial parameter sets generated by the method as suggested in section 2.2 of this paper. Random initial sets are randomly generated: probabilities are randomly drawn from the set $(0, 1)$; and trade rates are randomly drawn from the interval between the lowest and highest numbers of buys and sells with two restrictions. Pseudo-random initial sets are randomly generated after incorporating data properties. Explanation of the generation process is as presented in section 3.2. CL refers to initial parameter sets generated by our extension of the method of Cheng and Lai (2021). The numbers in the first column refer to the alternative numbers of parameter sets compared in each method. In each method and setting, the log-likelihood function in section 2.1 is maximized via MLE for each of the initial parameter sets and the maximization with the highest log-likelihood value is chosen. MAEs for PIN, PSOS as well as the probability variables $(\alpha, \delta, \theta, \theta')$ are in level differences between the estimated and actual values, while MAEs for the rate parameters $(\varepsilon_b, \varepsilon_s, \mu_b, \mu_s, \Delta_b, \Delta_s)$ are calculated in percentage differences between the estimated and actual values. All MAE numbers are reported in percentages for readability purpose. For example, in the first row, mean estimation error for PIN is 0.200% (in percentage units), and mean error for ε_b is 0.31% of actual ε_b (in percentage). The last column reports the mean running time in seconds. The results are for 1000 datasets simulated in accordance to the assumptions of the AdjPIN model. Data simulation as well as estimations are performed through the PINstimation package in R software. Total running time with an Í9 10990K processor is approximately 158 h.

via the GE method generate lower MAEs than estimations using 250 sets randomly (or pseudo-randomly) generated. Besides, the corresponding running times are around eightfold when running the estimations with 250 random sets. Table A2 presents p-values from the one-sided Wilcoxon signed rank test with an alternative hypothesis of 20 GE sets having lower MAEs than the ones of the alternative sets. The table shows that the use of 20 GE sets leads to MAEs in PIN and PSOS estimates that are significantly lower than the ones obtained by any other methods ($p < 0.01$). This result is same for all of the probability parameters and most of the intensity parameters in the model. In unreported analyses, we replicate the tables using mean squared errors, instead of mean absolute errors, obtaining qualitatively same results. Finally, the MAEs from the CL method based 256 sets reported in panel D of table 5 reveal that a superior estimation accuracy cannot be achieved despite the large number of sets and excessively long running times. MAEs for PIN and PSOS are substantially larger when compared to the ones from both 20 GE sets and 250 pseudo-random sets.

Having demonstrated the superior performance, and substantially low estimation errors for the GE method, we examine whether the performance also holds for restricted versions of the model. Instances of restricted models comprise, for instance, a model, with equal informed trading level in buy and sell sides ($\mu_b = \mu_s$) (M2 in table 6); or a model with identical buy side and sell side uninformed trading rates as well as equal probabilities of a liquidity shock on the information event days and on the no-information event days ($\varepsilon_b = \varepsilon_s$, and $\theta = \theta'$) (M10 in table 6).

Table 6 reports the MAEs for the estimation of 20 variants of the AdjPIN model, each using 20 GE initial parameter sets. We group the restricted models into three categories. First row in the table presents the results for the unrestricted model (M1) for comparison purpose. Panel A concerns models with restrictions on trading rate parameters, namely, informed trading rates (mu), uninformed trading rates (epsilon) and liquidity shock trading rates (delta) (models 2–7, M2–M7). We restrict one or several of these rates to be equal in the buy and sell sides. Panel B imposes an additional restriction on

Table 6. Mean absolute errors for estimations of the unrestricted and 20 restricted variants of the AdjPIN model—using 20 initial parameter sets generated by the GE method (GE-20).

	PIN	PSOS	α	δ	θ	θ'	ε_b	ε_s	μ_b	μ_s	Δ_b	Δ_s	Time
Unrestricted model													
M1	0.054	0.081	0.009	0.059	0.045	0.026	0.263	0.289	0.290	0.301	0.470	0.521	11.323
Panel A: Restrictions on intensity parameters													
M2	0.053	0.088	0.004	0.020	0.022	0.023	0.286	0.269	0.274	0.279	0.479	0.449	10.949
M3	0.050	0.075	0.004	0.040	0.013	0.032	0.199	0.199	0.282	0.280	0.445	0.463	10.980
M4	0.049	0.084	0.008	0.031	0.005	0.041	0.353	0.348	0.293	0.274	0.370	0.373	10.820
M5	0.052	0.088	0.001	0.002	0.022	0.003	0.365	0.356	0.265	0.273	0.394	0.394	10.302
M6	0.053	0.083	0.007	0.003	0.015	0.006	0.223	0.214	0.287	0.286	0.472	0.480	10.482
M7	0.055	0.082	0.001	0.001	0.002	0.001	0.218	0.221	0.276	0.286	0.379	0.379	9.725
Panel B: Restrictions on intensity parameters and/or $\theta = \theta'$													
M8	0.053	0.074	0.015	0.035	6.339	5.167	0.263	0.287	0.303	0.404	0.497	0.463	10.563
M9	0.047	0.069	0.003	0.006	5.962	5.434	0.288	0.278	0.260	0.262	0.422	0.403	10.437
M10	0.072	0.085	0.092	0.015	6.173	5.496	0.206	0.212	0.296	0.280	0.393	0.408	10.218
M11	0.051	0.069	0.001	0.001	5.414	5.839	0.335	0.338	0.301	0.301	0.298	0.299	10.209
M12	0.053	0.070	0.001	0.002	6.220	5.659	0.335	0.344	0.272	0.275	0.308	0.318	9.727
M13	0.053	0.073	0.005	0.004	6.324	5.319	0.231	0.235	0.275	0.287	0.429	0.437	9.713
M14	0.050	0.070	0.001	0.003	5.692	5.326	0.204	0.210	0.261	0.277	0.311	0.308	8.349
Panel C: Restrictions on intensity parameters and/or $\theta' = 0$													
M15	0.461	0.933	0.691	1.178	2.407	1.492	0.299	0.296	0.652	0.325	3.428	2.770	11.351
M16	0.509	0.948	0.800	0.953	2.271	1.452	0.269	0.274	0.234	0.240	1.985	2.898	11.322
M17	0.371	0.616	0.617	0.780	2.279	1.282	0.213	0.221	0.561	0.356	3.429	2.796	11.308
M18	0.081	0.122	0.084	0.195	0.287	0.062	0.234	0.242	0.116	0.125	0.156	0.160	10.887
M19	0.052	0.073	0.018	0.046	0.205	0.018	0.247	0.234	0.119	0.125	0.170	0.159	10.890
M20	0.246	0.536	0.339	0.462	1.792	0.919	0.195	0.198	0.168	0.173	2.017	1.684	11.011
M21	0.044	0.041	0.004	0.008	0.007	0.001	0.240	0.245	0.119	0.133	0.181	0.184	10.350

Notes: The table presents the mean absolute errors (MAEs) for adjusted probability of informed trading (PIN), probability of symmetric order flow shock (PSOS), and 10 intermediate parameters for each of the different AdjPIN models. First column presents the model name (M1–M21) as defined in table 7. For each of the 21 models, the log-likelihood function in section 2.1 is maximized via MLE for 20 sets of initial parameters obtained by the GE method suggested in section 2.2, and the maximization with the highest log-likelihood value is chosen. MAEs for PIN, PSOS as well as the probability variables $(\alpha, \delta, \theta, \theta')$ are in level differences between the estimated and actual values, while MAEs for the rate parameters $(\varepsilon_b, \varepsilon_s, \mu_b, \mu_s, \Delta_b, \Delta_s)$ are calculated in percentage differences between the estimated and actual values. All MAE numbers are reported in percentages for readability purpose. Last two columns report the mean running time (in seconds) and the number of converging maximizations. The results are for 21,000 datasets (1000 for each model) simulated based on the assumptions of the AdjPIN model and the included restrictions. For example, for the model with single uninformed intensity($\varepsilon_b = \varepsilon_s$), 1000 datasets are simulated with equal uninformed intensity in buy and sell sides. Data simulation as well as the estimations are conducted through the PINstimation package in R software. Total running time with an Í9 10990K processor is approximately 61 h.

Table 7. List of model restrictions.

M1: unrestricted	M8: $\theta = \theta'$	M15: $\theta' = 0$
M2: $\mu_b = \mu_s$	M9: $\mu_b = \mu_s, \theta = \theta'$	M16: $\mu_b = \mu_s, \theta' = 0$
M3: $\varepsilon_b = \varepsilon_s$	M10: $\varepsilon_b = \varepsilon_s, \theta = \theta'$	M17: $\varepsilon_b = \varepsilon_s, \theta' = 0$
M4: $\Delta_b = \Delta_s$	M11: $\Delta_b = \Delta_s, \theta = \theta'$	M18: $\Delta_b = \Delta_s, \theta' = 0$
M5: $\mu_b = \mu_s, \Delta_b = \Delta_s$	M12: $\mu_b = \mu_s, \Delta_b = \Delta_s, \theta = \theta'$	M19: $\mu_b = \mu_s, \Delta_b = \Delta_s, \theta' = 0$
M6: $\mu_b = \mu_s, \varepsilon_b = \varepsilon_s$	M13: $\mu_b = \mu_s, \varepsilon_b = \varepsilon_s, \theta = \theta'$	M20: $\mu_b = \mu_s, \varepsilon_b = \varepsilon_s, \theta' = 0$
M7: $\mu_b = \mu_s, \varepsilon_b = \varepsilon_s, \Delta_b = \Delta_s$	M14: $\mu_b = \mu_s, \varepsilon_b = \varepsilon_s, \Delta_b = \Delta_s, \theta = \theta'$	M21: $\mu_b = \mu_s, \varepsilon_b = \varepsilon_s, \Delta_b = \Delta_s, \theta' = 0$

the models M1–M7, which is the equality between the liquidity shock probability variables ($\theta = \theta'$) (M8–M14). Panel C replaces the restriction of $\theta = \theta'$ with a related one, setting the liquidity shock probability in information days variable equal to zero ($\theta' = 0$), implying that liquidity shocks only occur in no-information days (M15–M21). For each of the 21 models, we simulate 1000 datasets, 21,000 datasets in total. The list of restricted models in table 6 is reported below.

Mean absolute errors reported in table 6 are substantially small and stand as a supporting evidence for the consistency and reliability of the GE method suggested in this paper. The largest value among 240 reported MAEs (20 models, 10 intermediate parameters, PIN and PSOS) for M2 to M21 is 6.34%. Moreover, 83% of the reported MAEs are lower than 0.5%. 100% (78%) of the MAE in PIN and PSOS estimates are less than 1% (0.1%).

4. CONCLUSION

In this paper, we introduce an estimation method addressing the computational issues that plague in the estimation of the AdjPIN model of Duarte and Young (2009). While the model has been broadly used in the literature, associated computational problems have not received the necessary attention. We fill this gap in the literature by deriving the log-transformation of the likelihood function of the AdjPIN model and developing an initial parameter determination algorithm.

By the use of large number of simulated datasets with various characteristics, we provide evidence for the substantial decrease in the estimation biases towards zero. Estimations of the AdjPIN model using the original likelihood function do not converge when the maximum number of buys or sells in the data reaches to around 140. This implies that the likelihood function without the logarithmic transformation we present is not pertinent in today's financial markets. We also introduce a method (GE method) for obtaining strategically determined small number of parameter sets to initiate the maximization of the log-likelihood function. We show that using just 20 initial parameter sets provides a superior performance to the use of 250 randomly generated initial sets. Given that the use of a single or 10 randomly generated sets is the frequently used strategy in the literature, the paper has the potential to contribute. In fact, we provide supporting evidence on the estimation accuracy of the suggested method for 20 different variants of the AdjPIN model. Consequently, we strongly suggest the use of the introduced estimation method in further studies using the AdjPIN model. Due to the extreme ease of implementation of the method via the use of recently introduced R package PINstimation, we expect further studies to utilize the informed trading model more frequently, while benefiting from high estimation accuracy.

Acknowledgments

We thank participants at the Second Workshop on Market Microstructure and Behavioral Finance (WMMBF-II) and seminar participants at Kadir Has University International Trade and Finance Department, for the useful comments. We thank Duygu Atasever for her valuable research assistance.

Data availability statement

The data supporting the results and analyses presented in this paper have been simulated using the R package `PINstimation`. For further inquiries regarding the data or the simulation process, please contact the corresponding author.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Scientific and Technological Research Council of Turkey (TÜBÍTAK) [grant no 122K637].

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References

Cheng, T.C. and Lai, H.N., Improvements in estimating the probability of informed trading models. *Quant. Finance*, 2021, **21**(5), 771–796.

Dempster, A.P., Laird, N.M. and Rubin, D.B., Maximum likelihood from incomplete data via the EM algorithm. *J. R. Stat. Soc., B: Stat. Methodol.*, 1977, **39**(1), 1–22.

Duarte, J. and Young, L., Why is PIN priced? *J. Financ. Econ.*, 2009, **91**, 119–138.

Easley, D., Hvidkjaer, S. and O'hara, M., Is information risk a determinant of asset returns? *J. Finance*, 2002, **57**(5), 2185–2221.

Easley, D., Kiefer, N.M., Ohara, M. and Paperman, J.B., Liquidity, information, and infrequently traded stocks. *J. Finance*, 1996, **51**(4), 1405–1436.

Easley, D. and O'hara, M., Price, trade size, and information in securities markets. *J. Financ. Econ.*, 1987, **19**(1), 69–90.

Easley, D. and O'hara, M., Time and the process of security price adjustment. *J. Finance*, 1992, **47**(2), 577–605.

Easley, D. and O'hara, M., Information and the cost of capital. *J. Finance*, 2004, **59**(4), 1553–1583.

Ersan, O., Multilayer probability of informed trading. *SSRN Electron. J.*, 2016, **59**, 1553–1583.

Ersan, O. and Alici, A., An unbiased computation methodology for estimating the probability of informed trading (PIN). *J. Int. Financ. Mark. Inst. Money*, 2016, **43**, 74–94.

Gan, Q., Wei, W.C. and Johnstone, D., A faster estimation method for the probability of informed trading using hierarchical agglomerative clustering. *Quant. Finance*, 2015, **15**(11), 1805–1821.

Ghachem, M. and Ersan, O., PINstimation: an R package for estimating probability of informed trading models. *R. J.*, 2023, **15**(2), 145–168.

Ghachem, M. and Ersan, O., Estimation of the probability of informed trading models via an expectation-conditional maximization algorithm. *Financ. Innov.*, 2025, **11**(1), 67.

Lin, H.W. and Ke, W.C., A computing bias in estimating the probability of informed trading. *J. Financ. Mark.*, 2011, **14**(4), 625–640.

Yan, Y. and Zhang, S., An improved estimation method and empirical properties of the probability of informed trading. *J. Bank. Finance*, 2012, **36**(2), 454–467.

Appendix 1. Description of the data simulation process

The simulated datasets have been generated using the function `generate_adjpin()` from the package PINstimation (Ghachem and Ersan 2023). The function `generate_adjpin()` samples model parameters from the following default intervals: α, δ, θ, and θ' are drawn from the unit interval $(0, 1)$; ε_b is drawn from $(100, 10{,}000)$; ε_s from $(0.8 \times \varepsilon_b,\ 1.2 \times \varepsilon_b)$; and the liquidity shock intensities (Δ_b, Δ_s) and informed trading rates (μ_b, μ_s) are sampled from ranges dynamically determined based on ε_b and ε_s to ensure proper separation between the different day types.

The function then simulates 60 trading days, assigns each observation to one of six possible day types using the probabilities α, δ, θ, and θ', and generates buy and sell orders using Poisson distributions whose parameters are obtained from the generated trading levels. Empirical parameters are calculated and validated to ensure consistency with theoretical assumptions. This process ensures that the simulated datasets are representative of quarterly trading data. For further details regarding the function, consult the official documentation available at: https://pinstimation.com/reference/. The complete source code and implementation details can be accessed on the GitHub repository: https://github.com/monty-se/PINstimation.

Appendix 2. Descriptive statistics of the simulated datasets

Table A1. Descriptive statistics of the simulated datasets.

	Mean	sd	min	Q_{01}	Q_{05}	Q_{25}	median	Q_{75}	Q_{95}	Q_{99}	max
Buys	14,244.8	10,763.1	258.0	530.9	1378.4	5849.5	12,040.6	19,645.0	35,406.3	45,645.5	75,618.6
Sells	14,492.1	11,014.9	173.8	515.4	1407.6	5813.9	12,331.6	20,417.7	35,763.0	47,471.4	70,226.8
OI	− 247.3	6965	− 37,421	− 23,707	− 11,533	− 1905	− 41.9	1839.2	10,909	22,713.4	31,160
AOI	8089.9	8506.7	41.8	158.7	457.1	2063.5	5105.2	10,891.2	27,252.1	38,168.4	43,767.7
adjpin	0.252	0.160	0.006	0.015	0.037	0.114	0.226	0.367	0.543	0.614	0.688
psos	0.342	0.183	0.010	0.027	0.064	0.189	0.342	0.489	0.639	0.723	0.769
α	0.506	0.282	0.033	0.050	0.067	0.267	0.500	0.750	0.950	0.983	0.983
δ	0.511	0.286	0.017	0.020	0.057	0.261	0.505	0.760	0.950	0.977	0.983
θ	0.508	0.300	0.018	0.024	0.050	0.250	0.500	0.769	1.000	1.000	1.000
θ'	0.522	0.299	0.017	0.022	0.050	0.250	0.540	0.764	1.000	1.000	1.000
ε_b	5119.8	2972.6	124.2	211.5	581.8	2363.7	5161.1	7756.3	9617.9	9935.2	10,005.0
ε_s	5150.2	3063.8	112.5	205.8	586.7	2361.6	5129.8	7636.1	10,137.4	11,270.0	11,850.9
μ_b	14,990.7	12,255.8	116.4	496.1	1200.4	5115.6	11,625.0	21,737.6	40,578.8	48,531.7	57,350.3
μ_s	15,072.5	12,480.5	128.7	494.9	1181.8	5117.4	11,679.4	21,955.3	40,482.7	50,781.4	57,525.4
Δ_b	10,691.5	8889.9	82.2	244.7	763.7	3331.1	8125.5	16,341.8	28,573.8	34,991.7	39,567.7
Δ_s	10,659.3	8956.1	69.7	231.2	826.9	3336.2	8277.5	16,202.3	28,638.5	36,475.2	42,416.1

Notes: The table presents the descriptive statistics of the 1000 datasets used in table 5. Buys and Sell series are the mean number of buys and sells in each dataset. For example, mean statistic of sells, 14,492.1, is the value of 'dataset-level daily mean of sells'. OI and AOI are mean order imbalance and mean absolute order imbalance in each dataset. ADJPIN and PSOS are the average adjusted probability of informed trading and the probability of symmetric order flow shock. Q_x represents the *xth* quantile.

Appendix 3. Model comparisons

Table A2. Model comparisons: GE20 over random sets.

	PIN	PSOS	α	δ	θ	θ'	ε_b	ε_s	μ_b	μ_s	Δ_b	Δ_s
GE20 vs. Random.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
GE20 vs. Random.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.02	0.06
GE20 vs. Random.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.04	0.06	0.01	0.29	0.22
GE20 vs. Random.100	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.00	0.67	0.46
GE20 vs. Random.250	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.01	0.00	0.00	0.71	0.47
GE20 vs. PRandom.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.02	0.04
GE20 vs. PRandom.20	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.02	0.09	0.00	0.25	0.15
GE20 vs. PRandom.50	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.26	0.01	0.00	0.43	0.84
GE20 vs. PRandom.100	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.26	0.01	0.01	0.37	0.94
GE20 vs. PRandom.250	0.00	0.00	0.00	0.00	0.00	0.00	0.07	0.00	0.01	0.04	0.32	0.32
GE20 vs. CL	0.00	0.00	0.00	0.00	0.00	0.00	0.08	0.00	0.00	0.00	0.17	0.01

Notes: The table presents the p-values from the one-sided Wilcoxon signed rank test with an alternative hypothesis of 20 GE sets having lower MAEs than the ones of the alternative sets. The table is a supplement to table 5 where MAEs are reported. Columns represent adjusted probability of informed trading (PIN), probability of symmetric order flow shock (PSOS), and 10 intermediate parameters in the AdjPIN model. GE20 refers to 20 initial parameter sets generated by the method suggested in section 2.2 of this paper. Random.n refers to n randomly generated initial parameter sets: probabilities are randomly drawn from $(0, 1)$; and intensity rates are randomly drawn from the interval between lowest and highest numbers of buys and sells with two restrictions. PRandom.n refers to n randomly generated sets after incorporating data properties. Explanation of the generation process is as presented in section 3.2. CL refers to 256 initial parameter sets generated by our extension of the method of Cheng and Lai (2021) sets. In each setting, the log-likelihood function in section 2.1 is maximized via MLE for each of the initial parameter sets and the maximization with the highest log-likelihood value is chosen as optimal. The compared MAEs for PIN, PSOS as well as the probability variables $(\alpha, \delta, \theta, \theta')$ are in level differences between the estimated and actual values, while MAEs for the rate parameters (ε_b, ε_s, μ_b, μ_s, Δ_b, Δ_s) are calculated in percentage differences between the estimated and actual values. The results are for 1000 datasets simulated based on the assumptions of the AdjPIN model.